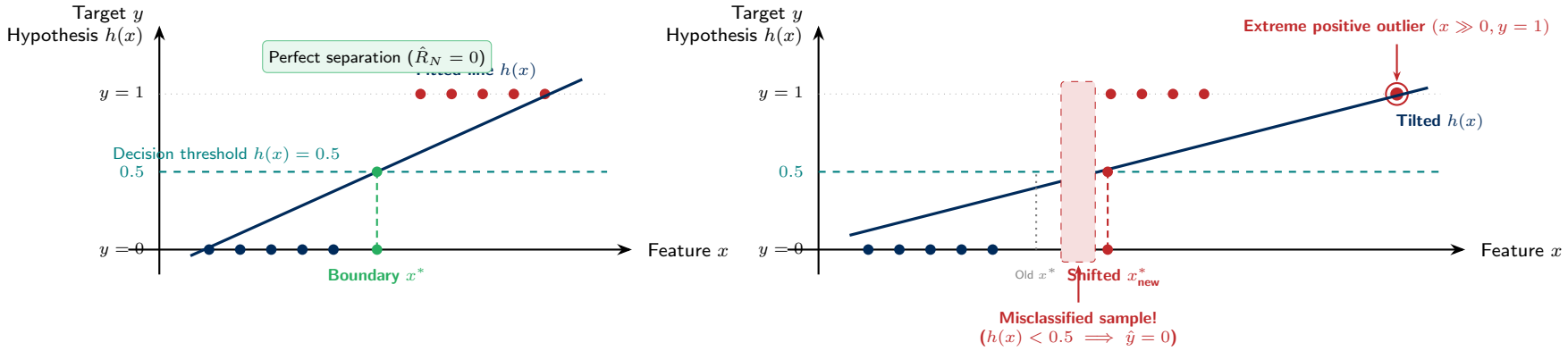

(a) Well-separated classes: least squares thresholding
Target y
Hypothesis $h(x)$
Perfect separation ($\hat{R}_N = 0$)
Fitted line $h(x)$
$y = 1$
Decision threshold $h(x) = 0.5$
0.5
$y = 0$
Boundary x^*
Feature x
(b) Outlier causes catastrophic decision boundary shift
Target y
Hypothesis $h(x)$
Extreme positive outlier ($x \gg 0, y = 1$)
Tilted $h(x)$
$y = 1$
0.5
$y = 0$
Old x^*
Shifted x^*_{new}
Misclassified sample!
($h(x) < 0.5 \implies \hat{y} = 0$)
Feature x